

Unit 9. Confidence Intervals and Hypothesis Testing

Inference on means and on the OLS slope

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lec. 19–23; OLS inference follows Stock & Watson Ch. 5.

Roadmap

- 1 Confidence intervals
- 2 Confidence intervals
- 3 Hypothesis testing
- 4 Duality and Neyman–Pearson
- 5 Inference on the OLS slope
- 6 Wrap-up

From point estimate to interval

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Definition: $1 - \alpha$ confidence interval

A random interval $[A(X), B(X)]$ such that

$$\mathbb{P}_{\theta_0}(A(X) \leq \theta_0 \leq B(X)) = 1 - \alpha \quad \text{for every } \theta_0.$$

Confidence intervals: the basic idea

A confidence interval measures the uncertainty around an estimate. Suppose we estimate an unknown parameter θ_0 using an estimator $\hat{\theta}$. Since $\hat{\theta}$ is computed from a random sample, it will generally not be exactly equal to θ_0 .

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For a 95% confidence interval, we usually write

$$\hat{\theta} \pm 1.96 \widehat{\text{SE}}(\hat{\theta}).$$

The number 1.96 comes from the standard normal distribution.

How to interpret a confidence interval

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If we repeatedly drew samples and constructed a confidence interval from each sample, then approximately 95% of those intervals would contain the true parameter θ_0 .

So the randomness is in the interval, not in the parameter.

Large-sample confidence intervals

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The t distribution has heavier tails than the normal distribution, so the confidence interval is wider.

As n becomes large, the t critical value becomes very close to the normal critical value.

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For a binary variable,

$$\text{Var}(\bar{X}_n) = \frac{p(1-p)}{n}.$$

The largest possible value of $p(1-p)$ is $1/4$, so

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A 95% confidence interval has approximate half-width

$$1.96 \times 0.0076 \approx 0.015.$$

So the margin of error is about 1.5 percentage points.

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We require

$$1.645 \frac{0.005}{\sqrt{n}} \leq 0.0005.$$

Solving for n gives

$$n \geq \left(\frac{1.645 \cdot 0.005}{0.0005} \right)^2 \approx 271.$$

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The important question is whether the standard error is appropriate for the problem:

- homoskedastic standard errors,
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- or other corrections.

So confidence intervals are only as reliable as the standard errors used to construct them.

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This is the “2 percentage point” margin of error reported in the press.

General recipe

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- 2 Find $a(\theta), b(\theta)$ with $\mathbb{P}_{\theta}(a(\theta) \leq \hat{\theta} \leq b(\theta)) = 1 - \alpha$.
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Example: $X_i \sim U[0, \theta]$, $\hat{\theta} = \max X_i$.

90% CI: $[\hat{\theta}/\sqrt[n]{0.95}, \hat{\theta}/\sqrt[n]{0.05}]$.

Notice: the CI depends on the data only through $\hat{\theta}$.

Hypothesis testing setup

Vocabulary

- **Null hypothesis** $H_0 : \theta \in \Theta_0$.
- **Alternative** $H_A : \theta \in \Theta_A$, disjoint from Θ_0 .
- **Test statistic** $T_n = T(X_1, \dots, X_n)$.
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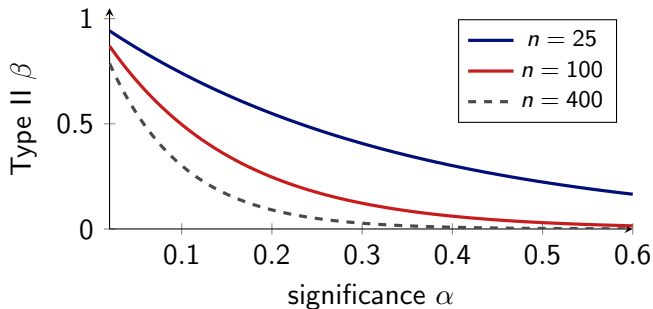
$\alpha = \mathbb{P}(\text{Type I})$; **Power** $1 - \beta = \mathbb{P}(\text{reject} \mid H_A)$.

The trade-off

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Convention: **control** α , then maximise power.

One-sided vs two-sided z -test

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One-sided. $H_A : \mu > \mu_0$: reject if $Z > z_{1-\alpha}$. At $\alpha = 0.05$: $Z > 1.645$.

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One-sided. $H_A : \mu > \mu_0$: reject if $Z > z_{1-\alpha}$. At $\alpha = 0.05$: $Z > 1.645$.

Unknown σ : use S_n , the statistic is $t = \sqrt{n}(\bar{X}_n - \mu_0)/S_n \sim t_{n-1}$ under H_0 .

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Why it matters

p -values quantify the strength of evidence: $p = 0.049$ is borderline, $p = 0.0001$ is overwhelming. Modern empirical practice reports p -values directly.

Duality between CI and test

Proposition

$$\theta_0 \notin \text{CI}_{1-\alpha}(X) \iff \text{reject } H_0 : \theta = \theta_0 \text{ at level } \alpha.$$

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Practical reading.

- A 95% CI gives, for free, a 5% test against **every** value outside it.
- Conversely, the **set of θ_0 that the test does not reject** is a $1 - \alpha$ confidence set.

Sampling distribution of $\hat{\beta}_1$

Under (LSA1)–(LSA3) (recall Unit 9):

Theorem

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Take-away

All inference on β_1 is just a CLT-based z-test. The framework of this unit applies directly.

Confidence interval and t -test on β_1

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Default in regression output: $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$, asking whether X helps predict Y at all.

Worked example: bus commute, continued

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Test $H_0 : \beta_1 = 0$. $t = 6/1.5 = 4.0$, $p \approx 6 \times 10^{-5} \Rightarrow$ strongly significant. **Test $H_0 : \beta_1 = 5$.**

$t = (6 - 5)/1.5 = 0.67$, $p \approx 0.50 \Rightarrow$ do not reject.

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The data are consistent with each bus adding 5 minutes.

Joint hypotheses: the F -test

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which is $\sim \chi_q^2/q$ asymptotically under H_0 .

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Default in regression software: F -test of **overall significance**, $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$.

Take-aways

- 1 A confidence interval = estimate $\pm$ critical value $\times$ standard error. Four cases by the distribution of $\hat{\theta}$.
- 2 A hypothesis test controls Type I error at α and seeks high power. The p -value summarises the evidence.
- 3 **Duality**: CIs and tests are two views of the same object.
- 4 **Neyman–Pearson**: the likelihood ratio test is most powerful for simple-vs-simple hypotheses.
- 5 **OLS inference**: t -statistics and CIs for $\hat{\beta}_1$ are built directly from the asymptotic normality of $\hat{\beta}_1$; F -tests handle joint hypotheses.

End of the course

You now have everything you need to start an econometrics course: random variables $\rightarrow$ regression $\rightarrow$ inference. Welcome to the econometrics journey.

Thank you

Questions?

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